

Mohammad Abdul Hadi

LLM Architecture · Security & Optimization · Agent Design · MCP Orchestration

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PROFESSIONAL SUMMARY

AI Security Researcher (Level: Sr. SDE) at **Huawei R&D**, operating at the intersection of **Malware analysis** (LLM for Security) and **Adversarial Machine Learning** (Security for LLM). Deep expertise in *LLM architecture optimization, Transfer Learning, Multi-agent and Tool Orchestration, and hardware-aware scaling*. Proven track record of optimizing AI infrastructure, achieving **315% performance gains** in production environments. Lead architect and developer of a **Agentic LLM ecosystem with 15+ agents and tools** for zero-day malware analysis, deployed across diverse hardware (NPU, GPU clusters, and IoT edge). Published researcher (150+ citations across A* and A rated Conferences).

EXPERIENCE

Huawei Technologies Canada Co., Ltd.

AI Security Researcher | Anshi Lab (Level: Senior Software Engineer)

Sep 2022 – Present

Vancouver/Burnaby, BC

► Project: Semantic LLM — Comprehensive Binary Analysis for Malware Detection

- **Hierarchical Transformer model: Recast Self-Attention with Vector Symbolic Architecture (VSA)** for advanced binary malware analysis, modernizing foundational program analysis literature.
Tech Stack: Python, C++, Assembly, PyTorch, Hugging Face (Transformers)
- **Semantic Function Model (SFM):** Developed two variants with respect to architecture dependency: **(a) Tokenless Instruction Set Transformer** – takes 32d *architecture specific* instruction-set as input and dispenses the need for a separate tokenizer; **(b) Intermediate Representation Tokenizer** – takes lifted LLVM IR with a **POV Normalization Engine** to achieve *architecture-agnostic* analysis.
Tech Stack: Python, C++, Assembly, PyTorch, Hugging Face (Transformers)
- **Semantic Program Model (SPM):** Swapped *Self-Attention* with **Holographic Reduced Representations** in a transformer. This architecture maps XOR-logic to the Query-Key interaction, with $\mathcal{O}(T \log T)$ complexity, handling malicious binaries with **100,000+ functions**. This is inherently resistant to adversarial attacks because the symbolic binding operations act as a natural 'clean-up' filter for input noise.
Tech Stack: Python, PyTorch, Hugging Face (Transformers) Python, PyTorch, Pinecone
- **Malware Analyst LLM:** Utilized **Mixture-of-Experts (MoE) routing** across SFM and SPM pathways to build an inclusive model. Elevated it's container framework through **Agent Client Protocol (ACP)** and **Model Context Protocol (MCP)** infrastructure that dynamically coordinate **6 multi-turn Online Agents and 10+ autonomous tools**. **Tools include:** Ghidra Pro Disassembler, LLVM IR Lifter, Static and Dynamic Behavior Logger (*Emulator*), and Chroma RAG-DB, etc. **Agents include:** Program Encoder *Signature Generator*, KNN Search, CFG Segment Classifier (*GAT*), LLM4Decompile Code Generation, and Pangu-R1 reasoning for explainability, etc.
Tech Stack: Python, Multi-Agent Systems, MCP, DSPy, PydanticAI, CrewAI, Pinecone
- **Heterogeneous-Hardware Deployment:** Performed **W8A8 dynamic quantization using Huawei's proprietary HiFloat (HF8)** via CANN & ModelSlim for Huawei NPU; Progressive Teacher-Student Distillation for IoT/edge; mixed-precision training for GPU clusters against open-source and in-house malware repositories.
Tech Stack: Python, C++, PyTorch, Hugging Face (PEFT), LoRA/QLoRA, CANN, vLLM, NVIDIA TensorRT-LLM, CPU/GPU/NPU Profiling
- **Scaled Downstream Capabilities:** **Function DNA Matching**, **vulnerability auditing**, and **cross-architecture code similarity search** using high-dimensional function/program embeddings generated by SFM and SPM.
Tech Stack: Python, PyTorch, Pinecone

► Project: High-Throughput ACL Pattern Matching via GPU-Accelerated Convolution

- **GPU-accelerated ACL Pattern Matching Engine:** Treats each Access Control rule as a 5-d filter and apply a **convolutional linear scan** over millions of rules for incoming packets in parallel on Tesla V100.
Tech Stack: C, C++, PyTorch, Linux Kernel
- **Structure-of-Arrays (SoA) for Rule Representation:** Proposed a novel and compact representation (26 bytes/rule), enabling coalesced memory access across threads; system stores up to **5M rules** within 124 MB of GPU memory.
Tech Stack: C, C++, CPython

- **Throughput Optimization:** Achieved kernel throughput of **100M packets/second** at 1K rules, sustaining ~80K packets/s at 5M rules; early-termination on first full match ($S(p, r) = 5$) minimizes wasted computation.
Tech Stack: C, C++, Linux Kernel
- **GPU profiler:** Collects GPU spec on-the-fly and sweeps for block and batch sizes, delivering up to **1.5× total-time speedup** for the 5M-rule case by reducing kernel time ~30% via optimal batch partitioning.
Tech Stack: C, C++, CPU/GPU Profiling

▷ *Project: Malware Filter Framework (MFF)*

- **Atrous Spatial Pyramid Pooling:** Optimized simple CNN in MFF by implementing dilation, achieving a **315% performance boost** via feature-profiling, memory caching, and multi-view filtering.
Tech Stack: C, TensorFlow
- **Model Lifecycle Monitoring and Maintenance Tools Incorporation:** Drove module refactoring and introduced industry-leading testing and software build practices to improve codebase quality and engineering efficiency.
Tech Stack: CI/CD
- **Technology map:** Linux C, user-space process development, kernel module development, memory allocation optimization, and low-level performance instrumentation.
Tech Stack: C, Linux Kernel

▷ *LLMOps & AI Research Leadership*

- **LLMOps Pipelines Management:** Directed multiple lifecycle components, including Model Versioners, Validators, Regression Testing, Runtimes, Schedulers, Domain/Data-Drift Detectors, and Retrainers.
Tech Stack: Python, Go, MLflow, Weights & Biases, Logfire, LangSmith, PromptLayer, Docker, Kubernetes, Temporal, Apache Airflow, Ray, GCP (Vertex, GKE)
- **PatentLM:** Proposed a dedicated **fine-tuned LLM** trained on Huawei's massive patent repository to search and align proprietary innovations, maximize R&D resource utilization, and streamline cross-domain research collaboration.
Tech Stack: Python, OpenAI, Anthropic, Gemini, Ollama, RLHF, DPO, LangChain, Weaviate
- **Champion Team Cohesion:** Brainstorm with malware analysts in every quarter to scope LLM solutions for detection and triage; drive cross-functional cybersecurity research and report findings to stakeholders.
Tech Stack: LangSmith, PromptLayer
- **Utilizing In-House Resources:** Enable large-scale training and inference of AI models on **novel Huawei AI hardware** (Ascend NPU); written lower-level APIs to support modern architectures across purpose-built AI systems.
Tech Stack: Python, Go, Mindspore, Docker, Kubernetes, Ray, FastAPI, gRPC

MineSense Technologies Ltd.

Firmware Engineer (Contract)

Jan 2023 - Dec 2023

Vancouver, BC – Hybrid

- **SLAM based Multi-Robot Communication & Motion Control Suite:** Developed and maintained embedded C firmware for robot control; integrated SLAM sensor suites for navigation and obstacle avoidance; Designed motion/stability control algorithms and communication protocols for robot-to-robot and central system messaging.
Tech Stack: C, C++, Python, Swift (Mobile/iPad Controller UI), Linux Kernel
- **Regression Testing:** Validated reliability and safety through systematic testing and debugging cycles.
Tech Stack: Python, CI/CD

Singapore Management University (SOAR Lab)

Research Assistant — DeepSense Project

Sep 2021 – Apr 2022

Singapore / On-Site

- **DeepHarvest:** LLM to collect and cross-link API information from GitHub, Stack Overflow, and Jira at scale.
Tech Stack: Python, SQL, Hugging Face Hub, GraphRAG, Elasticsearch
- **DeepRecommend:** API sequence recommendation engine supporting mixed natural-language and code queries.
Tech Stack: Python, SQL, RAG, LlamaIndex, pgvector (PostgreSQL)

Mitacs Internships: 3 (Scrawl / Niedner Inc. / Josef Schulte)

Software Engineer Intern

2020 – 2021

Canada

- **Scrawl:** Built plagiarism detection system for large-scale online document corpora; developed high-throughput NLP sentiment analysis pipelines.
Tech Stack: Python, TensorFlow, AWS (SageMaker), FastAPI

- **ARISE (Niedner/Josef-Schulte)**: Developed AI-driven production planning and management tools; interactive AI simulation tools for composite material modeling.

Tech Stack: Python, TypeScript, Hugging Face (Diffusers), AWS (EKS), FastAPI

The University of British Columbia

Graduate Research & Teaching Assistant

Sep 2019 – Mar 2023

- **Graduate Researcher (Transfer Learning)**: Researched **Foundational LLMs for Software Engineering**, including Code-Smell Detection, Code Comment Generation, and Code Synthesis using pre trained transformer models.
- **Graduate Teaching Assistant**: Parallel Computing, Database Systems, Computer Networking, and Human-Computer Interaction.
- **Software Developer Intern**: Developed interactive Web-based visualization tools for digital archaeological collections (Public Humanities Hub – Open Context platform).

Eastern University

Lecturer

Jul 2016 – Aug 2019

Dhaka, Bangladesh

- **Delivered 480+ Lectures spanning 10+ CS courses**: Mentored 600 students over 4 semesters.
- **Courses Taught**: Algorithm Design, Computer Architecture, Data Structures, Software Engineering, Digital Logic Design, Microprocessor/VLSI Design, Database Systems, and Web Frameworks.

EDUCATION

MSc in Computer Science (Thesis)

The University of British Columbia

2022

Summa Cum Laude

BSc in Computer Science & Engineering

North South University

2019

1st Position, Salutatorian, VC's Gold Medalist

TECHNICAL SKILLS

Languages: C, C++, Python, Go, SQL, Assembly, Swift, TypeScript

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, Diffusers, PEFT), CANN, Mindspore

LLM Platforms & Serving: OpenAI, Anthropic, Gemini, vLLM, Ollama, NVIDIA TensorRT-LLM

LLM Technologies & Concepts: RAG, GraphRAG, Multi-Agent Systems, MCP, RLHF, DPO, DSPy, LoRA/QLoRA

Agent Frameworks & RAG: PydanticAI, LangChain, LlamaIndex, CrewAI

Vector Databases & Search: pgvector (PostgreSQL), Pinecone, Weaviate, Elasticsearch

MLOps & Observability: MLflow, Weights & Biases, Logfire, LangSmith, PromptLayer, Hugging Face Hub

Infrastructure & Cloud: Docker, Kubernetes, Temporal, Apache Airflow, Ray, GCP (Vertex, GKE), AWS (SageMaker, EKS)

Backend & Systems: FastAPI, gRPC, CI/CD, Linux Kernel, CPU/GPU/NPU Profiling

SELECTED PUBLICATIONS (ranked by citations · [Google Scholar](#) · h-index: 6 · citations: 150+)

- [A] **MA Hadi**, FH Fard. **Evaluating Pre-Trained Models for User Feedback Analysis in Software Engineering: A Study on Classification of App-Reviews**. *Empirical Software Engineering*, Springer, 28(4), 2023. [69 citations]
- [B] **MA Hadi**, FH Fard. **AOBTM: Adaptive Online Biterm Topic Modeling for Version-Sensitive Short-Texts Analysis**. *IEEE ICSME 2020*. [34 citations]
- [C] **MA Hadi**, INB Yusuf, F Thung, KG Luong, J Lingxiao, FH Fard, D Lo. **On the Effectiveness of Pretrained Models for API Learning**. *IEEE/ACM ICPC 2022*. [19 citations]
- [D] K Luong, **MA Hadi**, F Thung, F Fard, D Lo. **FACOS: Finding API Relevant Contents on Stack Overflow with Semantic and Syntactic Analysis**. *arXiv:2111.07238*, 2021. [7 citations]
- [E] K Luong, **MA Hadi**, F Thung, F Fard, D Lo. **ARSeek: Identifying API Resource Using Code and Discussion on Stack Overflow**. *IEEE/ACM ICPC 2022*. [6 citations]

AWARDS & SERVICE

Scholarships & Awards

- 2025 Huawei Future Star Award
- '25, '26 Huawei Golden Network Award
- UBC Graduate Fellowship (Top 5%)
- VC's Gold Medal (1st of 1,870 graduates)
- 100% Undergraduate Scholarship (Top 3%)

Community & Service

- Mentor: Speech-to-text LLM tools for medical history collection
- Supervisor: Provide Capstone Research and Thesis guidance for underprivileged youth from Bangladesh.
- Programming Workshop Designer: US Embassy Dhaka
- Student Volunteer: ICSE 2020 & ICSE 2021